# 6. Project Testing Phase

## 6.1 Introduction

Testing verifies that LegalEase satisfies the specified requirements and performs its functions correctly. It identifies errors, security issues, integration problems, and unexpected behavior. Each module is tested individually and then as an integrated system.

## 6.2 Objectives of Testing

- To verify that all requirements are implemented correctly.
- To identify and correct software errors.
- To ensure authentication works properly and protected data is inaccessible to unauthorized users.
- To verify that documents, clauses, and versions are stored and retrieved correctly.
- To evaluate the completeness, accuracy, and clarity of Gemini-generated drafts.
- To verify that exported PDF and Word files are correct and readable.
- To verify usability and reliability.

## 6.3 Testing Strategy

Unit Testing -> Integration Testing -> System Testing -> User Acceptance Testing

## 6.4 Unit Testing

Individual functions are tested in isolation, including authentication functions, form validation, prompt-building functions, draft JSON parsing, clause numbering, version handling, and database operations. For example, required fields such as party names and dates are checked before a draft is requested. Different input combinations, including empty input, very long input, invalid dates, and invalid JSON, are tested to confirm the expected result.

## 6.5 Integration Testing

- Frontend and backend communication.
- Backend and database operations.
- Authentication and protected APIs.
- Backend and Gemini model (draft generation).
- Backend and Gemini model (clause explanation and review).
- Editor and version saving.
- Document assembly and PDF / Word export.

## 6.6 System Testing

Complete workflows are tested: registration, login, choosing a document type, filling the questionnaire, generating a draft, editing a clause, regenerating a clause, explaining a clause, running a review, exporting the document, viewing history, and logging out.

## 6.7 Functional Test Cases

| Test ID | Function | Test Input / Action | Expected Result |
|---|---|---|---|
| FT01 | Registration | Enter valid user details | Account is created |
| FT02 | Registration | Enter incomplete details | Validation message appears |

| Test ID | Function | Test Input / Action | Expected Result |
|---|---|---|---|
| FT03 | Login | Enter valid credentials | User is logged in |
| FT04 | Login | Enter incorrect password | Login is rejected |
| FT05 | Document Selection | Choose a document type | Matching questionnaire is shown |
| FT06 | Input | Submit the form with required fields empty | Validation message appears |
| FT07 | Draft Generation | Submit complete details for a rental agreement | Structured draft with numbered clauses is generated |
| FT08 | Accuracy | Compare the draft with the entered details | Names, dates, and amounts match the input |
| FT09 | Placeholders | Leave an optional detail blank | A clear placeholder appears instead of an invented value |
| FT10 | Clause Editing | Edit a clause and save | Updated text is stored |
| FT11 | Regeneration | Regenerate one clause with extra instructions | Only that clause is replaced |
| FT12 | Explanation | Select a clause and request an explanation | Plain-language explanation is shown |
| FT13 | Review | Run a review on the draft | Missing clauses and risks are listed |
| FT14 | Export | Export as PDF and Word | Correct, readable files are downloaded |
| FT15 | History | Open history | Saved documents are displayed |
| FT16 | Logout | Select logout | Session ends |

## 6.8 Authentication and Security Testing

- Access to protected APIs without a token is rejected.
- Expired or invalid tokens are rejected.
- A user cannot view or edit another user's documents.
- Passwords are stored as hashes.
- The Gemini API key is not exposed to the browser.

## 6.9 AI Response Testing

Gemini outputs are evaluated for completeness, consistency with the user's inputs, clarity of language, clause structure, format validity, and safety using sample scenarios such as a non-disclosure agreement, a house rental agreement, a freelance contract, and an employment offer letter. Cases tested include empty or very long inputs, missing details, prompts that trigger safety blocks, requests for unlawful documents, and malformed JSON output. The application displays a notice that drafts are for information only, are not legal advice, and should be reviewed by a qualified lawyer before use.

## 6.10 Performance and Usability Testing

Page load time, draft generation time, explanation and review time, and export time are observed, along with behavior during Gemini delays. Usability is checked for navigation, clarity of questions and

messages, ease of editing, and responsiveness on desktop and mobile screens.

## 6.11 User Acceptance Testing

A small group of students, freelancers, and small business owners uses the application and provides feedback on ease of use, clarity of drafts, usefulness of explanations, and the export quality. Reported issues are corrected before the final demonstration.

## 6.12 Testing Summary

After testing and correction of defects, all major modules were found to operate according to the requirements, and the application was considered ready for demonstration.